

Task 1: News Event Data Collection, Cleaning, and Temporal Alignment with Bank Indonesia USD/IDR Data

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Abstract

This report describes the workflow used in Task 1 to collect, clean, filter, and temporally align geopolitical news-event data with Bank Indonesia USD/IDR exchange-rate observations. The pipeline uses structured GDELT Event data as the primary news source and applies chunk-based processing, date filtering, validity checks, deduplication, text normalization, numeric preprocessing, and timestamp standardization. The cleaned news events are then aligned with daily Bank Indonesia exchange-rate observations using Jakarta time (WIB), a 16:00 WIB cutoff, and explicit rules for weekends, holidays, and missing timestamps. The final workflow is designed to produce a consistent dataset suitable for subsequent analysis of the relationship between geopolitical news events and exchange-rate movements.

1 Reason for Choosing the News Source

The team selected the **GDELT (Global Database of Events, Language, and Tone)** Event dataset as the primary news-event source because it provides large-scale, structured information about events reported in news media. Rather than collecting articles individually from different websites, GDELT provides standardized event-level records containing information such as event identifiers, dates, actors, event codes, source URLs, and quantitative event-related measures.

The use of structured GDELT Event data also makes the data collection process more suitable for a large-scale data pipeline. The dataset used in this project contains millions

of records, allowing the analysis to cover a broad period while retaining information about individual geopolitical events.

The main reasons for selecting GDELT are:

- **Structured event representation:** Each observation contains standardized event-related fields such as `GlobalEventID`, `SQLDATE`, `DATEADDED`, and `SOURCEURL`.
- **Large coverage:** GDELT provides a very large number of event observations, which is appropriate for studying geopolitical events over an extended time period.
- **Machine-readable format:** The TSV format can be processed efficiently using Python and pandas.
- **Event-level information:** The dataset provides structured information about actors, event types, tone, mentions, sources, and articles.
- **Reproducibility:** Using a standardized event dataset makes the collection and preprocessing procedure easier to reproduce.

The analysis period in the pipeline is from **1 September 2021 to 1 September 2026**.

2 Scraping and Data Collection Technique

The pipeline does not scrape individual news webpages one by one. Instead, it processes the structured GDELT Event TSV dataset, which functions as the news-event data source.

The raw GDELT file is approximately 2.189 GB and contains 22 columns. Because the dataset is too large to efficiently load into memory as a single DataFrame, the pipeline uses **chunk-based reading**. Each chunk contains 100,000 rows.

The basic procedure is:

1. Load the GDELT Event TSV file in chunks of 100,000 rows.
2. Parse the relevant date fields.
3. Restrict records to the required analysis period.
4. Remove records without a valid event identifier or source URL.
5. Deduplicate event records.
6. Normalize textual fields.

7. Convert numerical fields to appropriate numeric types.
8. Parse timestamps and convert them to Jakarta time.
9. Prepare the cleaned event dataset for temporal alignment.

The use of chunk processing reduces memory requirements and allows the pipeline to process millions of observations without requiring the complete raw dataset to be loaded into memory simultaneously.

3 Filtering Strategy

Filtering is applied before the alignment stage to ensure that only relevant and usable observations are retained.

The main filtering rules are as follows.

3.1 Date Filtering

The `SQLDATE` field is parsed as a date and records outside the specified analysis period are removed. The selected period is:

$$\text{START_DATE} = 2021-09-01$$
$$\text{END_DATE} = 2026-09-01$$

This ensures that the news events correspond to the period covered by the analysis.

3.2 Validity Filtering

A news-event record is retained when it has at least one usable identifier between its event identifier and source URL. In particular, records are removed when both the event ID and source URL are empty or missing.

This prevents records with insufficient identifying information from entering the cleaned dataset.

3.3 Deduplication

Duplicate event records are removed using `GlobalEventID` as the primary identifier.

When an event identifier is unavailable, the pipeline uses a fallback composite key consisting of:

$$\text{source} \mid \text{actor1} \mid \text{event_code} \mid \text{actor2} \mid \text{event_date}$$

If multiple records represent the same event, the earliest `recorded_utc` observation is retained.

The filtering process resulted in approximately:

- Raw rows: **10,623,426**
- Rows after filtering: **10,601,190**
- Unique cleaned events: **10,601,190**

4 Data Cleaning and Preprocessing Strategy

After filtering, several preprocessing operations are performed to improve consistency and prepare the data for analysis.

4.1 Text Cleaning

Textual fields are first converted so that missing values do not cause processing errors. HTML-like tags are removed, whitespace is collapsed, and leading or trailing whitespace is stripped.

Conceptually, the text preprocessing follows:

raw text → fill missing values → remove HTML-like tags → normalize whitespace → trim

This produces cleaner and more consistent textual fields.

4.2 Numeric Cleaning

The following GDELT numerical variables are converted into appropriate numeric representations:

- `GoldsteinScale`
- `NumMentions`
- `NumSources`
- `NumArticles`
- `AvgTone`

This allows the fields to be used reliably in statistical analysis and aggregation.

4.3 Timestamp Processing

The `DATEADDED` field is interpreted as a UTC timestamp and converted to a timezone-aware datetime.

When a usable timestamp is unavailable, the pipeline falls back to `SQLDATE` at midnight UTC.

The resulting UTC timestamp is then converted to:

Asia/Jakarta (WIB)

This conversion is important because the corresponding Bank Indonesia observations are interpreted using Indonesian local time.

4.4 Bank Indonesia Data Cleaning

The Bank Indonesia exchange-rate data are cleaned into the following main variables:

- `date`
- `usd_idr`
- log return of USD/IDR

The exchange-rate log return is calculated as:

$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right)$$

where P_t represents the USD/IDR exchange rate on day t .

The final cleaned Bank Indonesia dataset contains approximately **1,202 observations**.

5 Alignment Rules for Timezone Shifts and Non-Trading Days

Temporal alignment is required because GDELT events are timestamped at the event-recording level, whereas Bank Indonesia exchange-rate data are organized by trading/observation dates.

The pipeline therefore converts all event timestamps to **Asia/Jakarta (WIB)** before determining the appropriate Bank Indonesia date.

5.1 Market Cutoff

A cutoff time of:

16:00 WIB

is used.

The alignment rules are:

1. If an event occurs at or before 16:00 WIB and the same date exists in the Bank Indonesia dataset, the event is assigned to the same date.
2. If an event occurs after 16:00 WIB, it is assigned to the **next available Bank Indonesia date**.
3. If an event occurs on a weekend or holiday for which no Bank Indonesia observation exists, it is assigned to the **next available Bank Indonesia date**.
4. If only an event date is available and no precise timestamp can be obtained, the pipeline uses the date-only fallback rule.
5. Events for which no valid Bank Indonesia observation can be found are excluded from the aligned dataset.

The alignment process also records an `alignment_reason` field. Possible categories include:

- `same_day_before_cutoff`
- `after_market_close`
- `weekend_or_holiday`
- `event_date_only_fallback`

This explicit labeling makes the alignment process auditable and allows future analysis to distinguish observations based on why they were assigned to a particular exchange-rate date.

6 Task 1 Workflow Diagram

Figure 1 provides a concise overview of the complete Task 1 workflow.

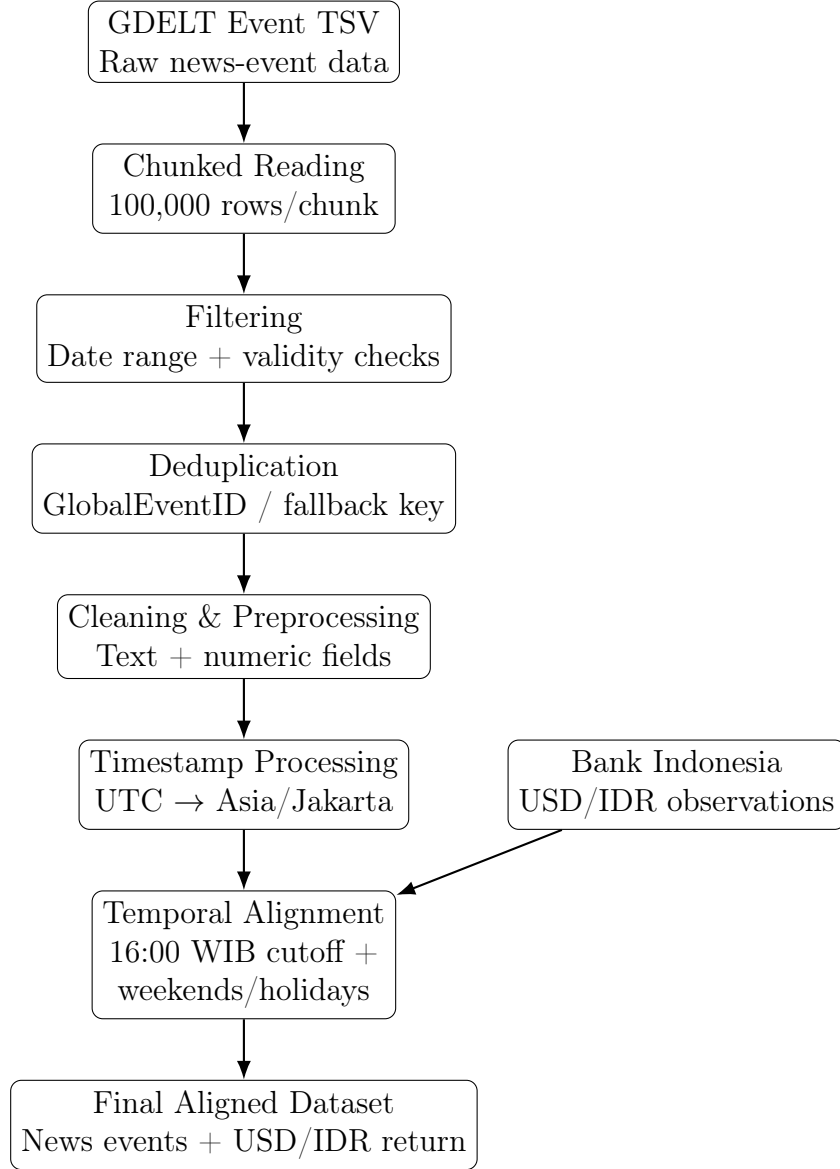


Figure 1: Overview of the Task 1 news-event processing and temporal alignment workflow.

7 Output and Validation

The pipeline generates cleaned and aligned datasets for subsequent analysis. Important output files include:

- `news_cleaned.parquet`
- `news_aligned.parquet`
- `daily_aligned_dataset.parquet`

The resulting aligned news dataset is approximately 911.97 MB, while the cleaned news dataset is approximately 910.72 MB.

The pipeline also performs validation checks covering:

- final date range,
- duplicate Bank Indonesia dates,
- missing USD/IDR observations,
- total number of event records,
- total number of unique source URLs, and
- distribution of alignment reasons.

These checks provide a final quality-control step before the data are used for downstream analysis.

8 Summary

The Task 1 pipeline establishes a reproducible process for transforming a large GDELT geopolitical event dataset into a cleaned and temporally aligned dataset with Bank Indonesia USD/IDR exchange-rate observations.

The main stages are:

1. Selecting structured GDELT Event data as the news source.
2. Reading the large dataset using memory-efficient chunk processing.
3. Filtering records by date and data validity.
4. Removing duplicate event observations.
5. Cleaning textual and numerical variables.

6. Standardizing timestamps from UTC to Jakarta time.
7. Cleaning Bank Indonesia USD/IDR observations and calculating log returns.
8. Aligning news events using a 16:00 WIB cutoff and explicit non-trading-day rules.
9. Validating and exporting the final aligned datasets.

This workflow provides a consistent foundation for examining the temporal relationship between geopolitical news events and movements in the USD/IDR exchange rate.

9 GitHub Repository

The complete source code, data-processing notebook, and related project materials are available in the team's GitHub repository.

<https://github.com/aufaarya/PROJECT-NLP.git>

Replace the URL above with the team's actual GitHub repository link.